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Problem 8.

$$1. f_X(x) = \begin{cases} \frac{1}{20}, & 0 \leq x \leq 20 \\ 0, & \text{otherwise} \end{cases} \quad \begin{array}{l} \text{When } 0 \leq X \leq 5, Y = 5 - X \\ \text{When } 5 < X \leq 20, Y = 20 - X \end{array}$$

$$P_Y(y) = \begin{cases} P_X(20-y), & 5 < y \leq 15 \\ \frac{y}{10}, & 0 \leq y \leq 5 \end{cases}$$

$$\text{if } 0 \leq y \leq 5, P_Y(y) = P(5-y \leq X \leq 5) + P(20-y \leq X \leq 5) \\ = \frac{y}{20} + \frac{y}{20} = \frac{y}{10}$$

$$\text{if } 5 < y < 15, P_Y(y) = P(20-y \leq X \leq 20) + P(0 \leq X \leq 5) \\ = \frac{y}{20} + \frac{5}{20} = \frac{y+5}{20}$$

$$f_Y(y) = \begin{cases} \frac{1}{10}, & 0 \leq y \leq 5 \\ \frac{1}{20}, & 5 < y \leq 15 \\ 0, & \text{otherwise} \end{cases}$$

$$2. \text{Denote } Z = X - Y, Z = X - Y$$

$$\text{When } z \geq 0, F_Z(z) = P(X - Y \leq z) = 1 - P(X - Y > z)$$

$$= 1 - \int_0^\infty dy \int_{y+z}^\infty \lambda e^{-\lambda x} \cdot \mu e^{-\mu y} dx = 1 - \int_0^\infty \mu e^{-\mu y} \cdot e^{-\lambda(y+z)} dy$$

$$= 1 - \int_0^\infty e^{-\lambda z} \cdot e^{-(\mu+\lambda)y} \cdot \mu dy$$

$$= 1 - \frac{e^{-\lambda z}}{\mu+\lambda} \cdot \mu = 1 - \frac{\mu}{\mu+\lambda} e^{-\lambda z}$$

$$\text{When } z < 0, F_Z(z) = P(X - Y \leq z) = P(Y - X \geq -z)$$

$$= \frac{\lambda}{\mu+\lambda} e^{-\mu(-z)} = \frac{\lambda}{\mu+\lambda} e^{\mu z}$$

$$F_Z(z) = \begin{cases} 1 - \frac{\mu}{\mu+\lambda} e^{-\lambda z}, & z \geq 0 \\ \frac{\lambda}{\mu+\lambda} e^{\mu z}, & z < 0 \end{cases}$$

$$f_Z(z) = \begin{cases} \frac{\lambda \mu}{\mu+\lambda} e^{-\lambda z}, & z \geq 0 \\ \frac{\lambda \mu}{\mu+\lambda} e^{\mu z}, & z < 0 \end{cases}$$

$$3. f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$f_Y(y) = f_X(\log y) \cdot \frac{d}{dy}(\log y) = \frac{1}{y\sigma\sqrt{2\pi}} e^{-\frac{(\log y - \mu)^2}{2\sigma^2}}$$

$$4. f_{X,Y}(x,y) = \frac{1}{\pi x^2} = \frac{1}{\pi}$$

$$F_R(r) = P(R \leq r) = P(X^2 + Y^2 \leq r^2) = \frac{\pi r^2}{\pi} = r^2, \quad f_R(r) = \frac{dF_R(r)}{dr} = 2r$$

$$F_X(x) = \int_{-1}^x \int_{-\sqrt{1-\xi^2}}^{\sqrt{1-\xi^2}} \frac{1}{\pi} dy d\xi = \frac{2}{\pi} \int_{-1}^x \sqrt{1-\xi^2} d\xi$$

$$\text{令 } \xi = \cos \theta, \theta \in [0, \pi], \quad d\xi = -\sin \theta d\theta$$

$$\frac{2}{\pi} \int_{-1}^x \sqrt{1-\xi^2} d\xi = \frac{2}{\pi} \int_{\pi}^{\arccos x} \sin \theta \cdot (-\sin \theta) d\theta = \frac{1}{\pi} \int_{\arccos x}^{\pi} (1 - \cos 2\theta) d\theta$$

$$= \frac{\pi - \arccos x}{\pi} - \frac{1}{2} \sin 2\theta \Big|_{\arccos x}^{\pi} = 1 - \frac{\arccos x}{\pi} + \frac{1}{2\pi} \sin(2 \arccos x)$$

$$= 1 - \frac{\arccos x}{\pi} + \frac{1}{\pi} \sqrt{1-x^2} \cdot x$$

$$f_X(x) = \frac{1}{\pi} \left(\frac{1}{\sqrt{1-x^2}} \right) + \frac{1}{\pi} \left(\frac{-2x^2}{2\sqrt{1-x^2}} + \sqrt{1-x^2} \right) = \frac{1}{\pi} \cdot \frac{2-2x^2+2-2x^2}{2\sqrt{1-x^2}} = \frac{4}{\pi} \frac{\sqrt{1-x^2}}{2} = \frac{2}{\pi} \sqrt{1-x^2}$$